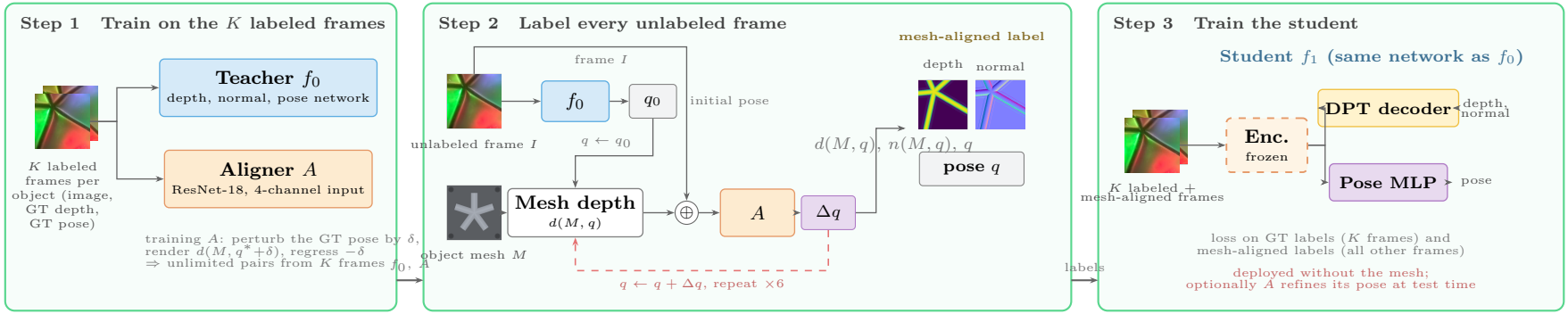

Step 1 Train on the K labeled frames
Teacher f_0
depth, normal, pose network
Aligner A
ResNet-18, 4-channel input
K labeled frames per object (image, GT depth, GT pose)
training A: perturb the GT pose by δ, render $d(M, q^*+\delta)$, regress $-\delta$
⇒ unlimited pairs from K frames
f_0, A
Step 2 Label every unlabeled frame
mesh-aligned label
frame I
depth
normal
f_0
q_0
initial pose
unlabeled frame I
$q \leftarrow q_0$
$d(M, q)$, $n(M, q)$, q
pose q
Mesh depth
$d(M, q)$
A
Δq
object mesh M
$q \leftarrow q + \Delta q$, repeat ×6
labels
Step 3 Train the student
Student f_1 (same network as f_0)
DPT decoder
depth, normal
Enc.
frozen
Pose MLP
pose
K labeled + mesh-aligned frames
loss on GT labels (K frames) and mesh-aligned labels (all other frames)
deployed without the mesh;
optionally A refines its pose at test time